

Enhancing Digital Government & Economy (EDGE) Project

Institute of Information & Communication Technology

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Digital Skills for Students Program



Course Title: Full Stack Web Development with React and Express.js

Batch – FSWD-01

Final Project on

“BookVerse Full Stack MERN Project”

Submitted To

Md. Mortuza Hossain & Md. Mosabbir Hossain

Trainer, EDGE Project

IICT, DUET

Submitted By

117	Mina Sarkar Nishi	119	Jayed Hoshen
118	Md. Sabbir Ahmed Khan	120	Mizanur Rahman

BookVerse

Professional MERN Stack Online Book Store Project Report

Project Type	Full Stack MERN Application
Frontend	React.js + Tailwind CSS + DaisyUI
Backend	Node.js + Express.js
Database	MongoDB Atlas
Authentication	JWT + Protected Routes

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1. Introduction

BookVerse is a modern full-stack online bookstore and reading platform developed using the MERN stack architecture. The application was designed to provide users with a premium and elegant digital reading experience inspired by editorial and literary publishing platforms.

The project focuses on clean typography, responsive layouts, dynamic catalog browsing, secure authentication systems, and scalable backend architecture. Users can browse books, filter collections by category, add products to cart, manage their reading activities, and access a personalized dashboard.

BookVerse combines modern frontend technologies with secure backend APIs to create a complete real-world eCommerce solution. The system was developed with performance, maintainability, and user experience in mind.

2. Project Objectives

The major objectives of the BookVerse project are:

- To create a professional online bookstore platform.
- To implement secure authentication and protected routes.
- To develop a responsive editorial-inspired user interface.
- To create scalable REST APIs using Express.js.
- To manage books, reviews, orders, and users dynamically.
- To implement modern frontend state management.
- To gain real-world MERN stack development experience.
- To improve full-stack architectural understanding.
- To practice frontend-backend integration professionally.

3. System Overview

BookVerse consists of two primary systems: frontend and backend.

Frontend: The frontend is built using React.js with Vite for fast development and optimized performance. Tailwind CSS and DaisyUI were used to design a luxury editorial-inspired user interface.

Backend: The backend is developed using Node.js and Express.js. MongoDB Atlas serves as the primary cloud database. RESTful APIs are used for frontend-backend communication.

Main Modules:

- Authentication Module
- Book Management Module
- Order Management Module
- Cart System
- Review System
- User Dashboard
- Admin Dashboard

4. Technology Stack

Frontend Technologies:

- React.js
- Vite
- Tailwind CSS
- DaisyUI
- React Router DOM
- React Query
- Axios
- Lucide React

Backend Technologies:

- Node.js
- Express.js
- MongoDB Atlas
- Mongoose
- JWT Authentication
- bcryptjs
- cookie-parser
- CORS

Development Tools:

- Git & GitHub
- VS Code
- Netlify Deployment
- Render Backend Hosting

5. Backend Development

The backend of BookVerse follows a modular and scalable architecture.

Folder Structure:

```
/src  
  /controllers  
  /models  
  /routes  
  /middlewares  
  /services  
  /utils
```

Backend Features:

- User Authentication
- Role-Based Authorization
- CRUD Operations for Books
- Cart Management
- Order Processing

- Review Management
- Dynamic Filtering
- Secure Middleware Validation

The backend APIs are structured professionally to support scalable future development.

6. Frontend Development

The frontend was designed with modern editorial aesthetics to create a luxury reading experience.

Frontend Features:

- Responsive Navbar
- Hero Section
- Dynamic Catalog Page
- Trending Books Section
- Book Details Page
- Interactive Dashboard
- Elegant Login/Register Pages
- Dynamic Cart Page
- Protected Routes

React Query is used for intelligent data fetching and caching. Reusable components help maintain a clean codebase and scalable architecture.

7. Authentication & Authorization

BookVerse uses JWT-based authentication for secure access control.

Authentication Flow:

1. User registers or logs in.
2. Backend validates credentials.
3. JWT token is generated.
4. Protected routes verify the token.
5. Users gain access to private pages.

Authorization Levels:

- Guest User
- Authenticated User
- Admin User

Security Features:

- Password Hashing
- Protected APIs
- Role-Based Access
- Secure Logout System

8. Database Design

MongoDB Atlas is used as the primary cloud database.

Main Collections:

Users Collection:

- Name
- Email
- Password
- Role
- Reading History

Books Collection:

- Title
- Author
- Genre
- Price
- Stock
- Description
- Cover Image

Orders Collection:

- User Information
- Ordered Books
- Shipping Address
- Order Status

Reviews Collection:

- User ID
- Book ID
- Rating
- Comment

The database schema was designed for scalability and efficient querying.

9. API Integration

BookVerse uses Axios and React Query for API communication.

Major APIs:

- Authentication API
- Books API
- Orders API
- Reviews API
- Cart API

Benefits:

- Fast synchronization
- Better caching
- Cleaner frontend logic
- Optimized performance

REST APIs were designed to follow scalable and maintainable standards.

10. UI/UX Design Philosophy

The BookVerse design philosophy is inspired by premium literary and publishing platforms.

Design Characteristics:

- Minimalistic Layouts
- Serif Typography
- Soft Neutral Colors
- Editorial Grid Structure
- Responsive Design
- Luxury Dashboard UI
- Smooth User Experience

The interface focuses on readability and immersive browsing experiences.

11. Main Features

User Features:

- User Registration & Login
- Dynamic Book Browsing
- Search System
- Genre Filtering
- Add to Cart
- Reading Dashboard
- Reviews & Ratings
- Order History

Admin Features:

- Dashboard Analytics
- Product Management
- User Monitoring
- Order Management
- Review Moderation

Additional Features:

- Protected Routes
- Responsive UI
- Dynamic Navigation
- Real-time API Data

12. Security Implementation

Several security mechanisms were implemented in BookVerse.

Security Practices:

- JWT Authentication
- Password Hashing using bcrypt
- Protected Routes
- Role-Based Authorization
- Secure API Access
- Input Validation
- CORS Configuration

These measures ensure secure communication between frontend and backend systems.

13. Performance Optimization

Frontend Optimization:

- React Query Caching
- Reusable Components
- Optimized Rendering
- Lazy Loading
- Fast Vite Build System

Backend Optimization:

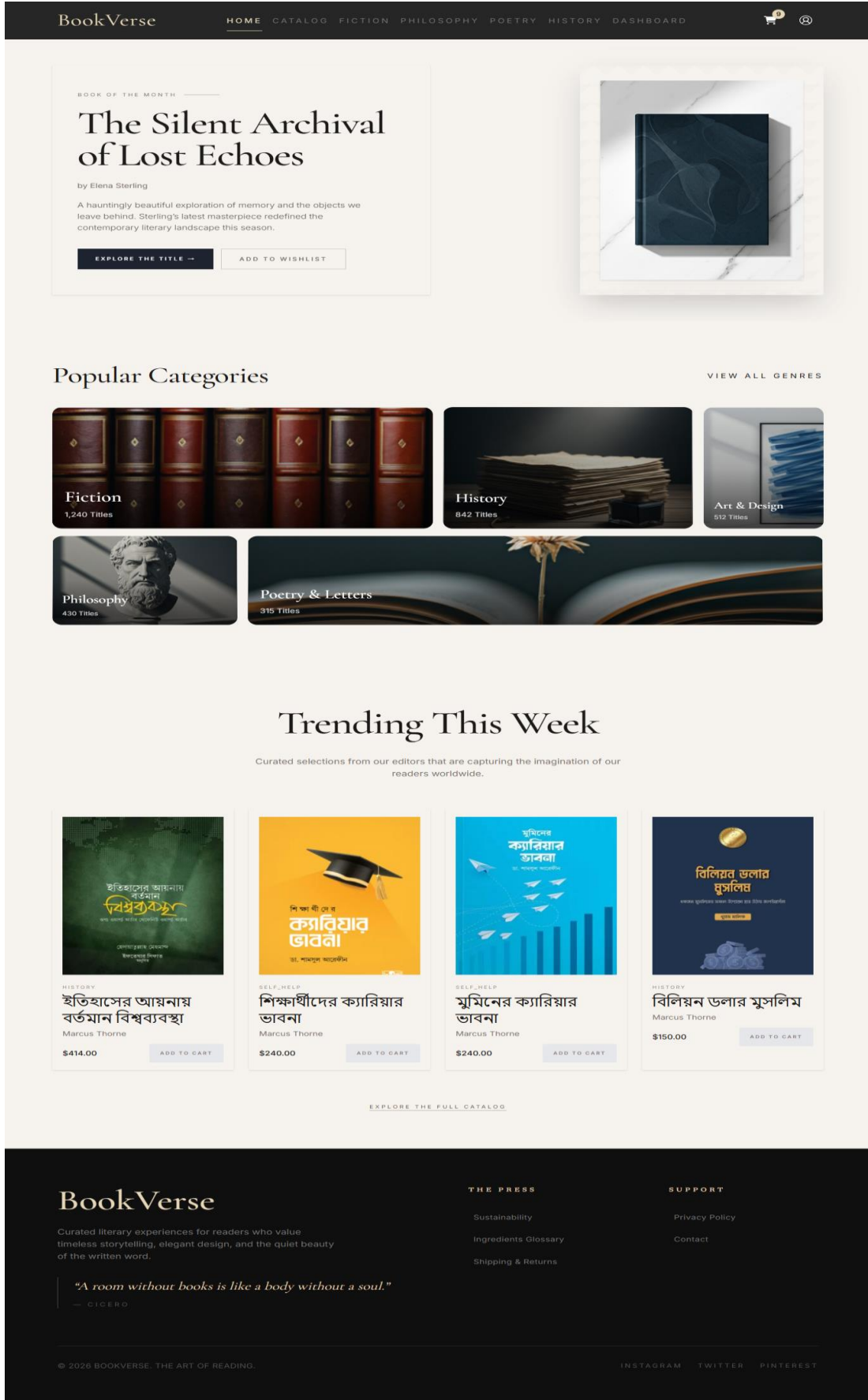
- Efficient MongoDB Queries
- Middleware Validation
- Modular Code Architecture
- Structured API Responses

The system was optimized for smooth performance across desktop and mobile devices.

14. Main Pages & Screenshot Sections

The following sections are reserved for project screenshots.

Home Page



Catalog Page

BookVerse

HOME CATALOG FICTION PHILOSOPHY POETRY HISTORY DASHBOARD

🛒

👤

Curated Catalog

Showing 1 titles from independent presses

GENRE

☐ Contemporary Fiction

☐ Classic Literature

☐ Philosophy & Essays

☐ Poetry

☐ Art & Design

AUTHORS

Joan Didion

Albert Camus

Virginia Woolf

PRICE RANGE

\$10\$150

CLEAR ALL FILTERS

বিলিয়ন ডলার মুসলিম

৳১৫০

বিলিয়ন ডলার মুসলিম

Hafez Jayed

\$150

HISTORY

Book Details Page

BookVerse

HOME CATALOG FICTION PHILOSOPHY POETRY HISTORY DASHBOARD

🛒

👤

বিলিয়ন ডলার মুসলিম

৳১৫০

বিলিয়ন ডলার মুসলিম

Hafez Jayed

\$150

HISTORY

বিলিয়ন ডলার মুসলিম

\$150

৳১৫০

ADD TO CART

Free shipping on orders over \$75

Key Actives

No active ingredients listed.

The Ritual

Ritual steps coming soon.

Login Page

[← BACK HOME](#)

BookVerse

CURATED FOR THE BIBLIOPHILE

Welcome Back

Enter your details to continue your literary journey through BookVerse.

EMAIL ADDRESS

user1@gmail.com

PASSWORD

FORGOT PASSWORD?

.....

☐ Remember me for 30 days

SIGN IN

Don't have an account? [Register](#)

BOOKVERSE

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Register Page

[← BACK HOME](#)

BookVerse

BEGIN YOUR LITERARY JOURNEY

Create Account

Join BookVerse and build your personal collection of timeless literary works.

USERNAME

e.g. literarysoul

EMAIL ADDRESS

reader@bookverse.com

PASSWORD

MINIMUM 8 CHARACTERS

.....

☐ I agree to the Terms of Service and Privacy Policy.

CREATE ACCOUNT

Already have an account? [Sign In](#)

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User Dashboard Page

BookVerse

HOME CATALOG FICTION PHILOSOPHY POETRY HISTORY DASHBOARD

9

User1

Premium Bibliophile

WELCOME BACK

Reading Dashboard

Overview

Profile

Order History

My Library

Reading Progress

Analytics

My Reviews

CURRENT READING

Literary Journey

View Full Library

milk and honey

memoir

ACTIVE NOW

The Architecture of Happiness

by Alain de Botton

Explore timeless perspectives and ideas through your curated reading experience on BookVerse.

Reading Progress

68%

CONTINUE READING

PURCHASE ACTIVITY

Recent Orders

RECOMMENDED READ

SEO

THE 4-HOUR WORKWEEK

THE 5000 PAGES OF THE HISTORY OF THE WORLD

THE ART OF THINKING

THE ART OF WRITING

THE ART OF THINKING

THE ART OF WRITING

THE ART OF THINKING

THE ART OF WRITING

Meditations

Marcus Aurelius

"The happiness of your life depends upon the quality of your thoughts."

PERSONAL NOTES

My Reviews

Write Review

The Overstory

2 days ago

A magnificent and deeply reflective narrative about interconnected lives, nature, and human emotion.

Circe

1 week ago

Beautifully written, emotionally rich, and unforgettable in its storytelling and atmosphere.

BookVerse

Curated literary experiences for readers who value timeless storytelling, elegant design, and the quiet beauty of the written word.

"A room without books is like a body without a soul."

— CICERO

THE PRESS

Sustainability

Ingredients Glossary

Shipping & Returns

SUPPORT

Privacy Policy

Contact

© 2026 BOOKVERSE. THE ART OF READING.

INSTAGRAM TWITTER PINTEREST

Admin Dashboard

BookVerse

Admin Dashboard

Sign Out

Analytics

Inventory

Orders

Customers

MANAGEMENT OVERVIEW

Welcome back, admin.

+ Add New Book

Curate the collection, monitor user engagement, and oversee the library flow of the BookVerse ecosystem.

TOTAL BOOKS

4

Fresh titles in inventory.

ACTIVE USERS

2

Currently active library members.

REVENUE (MTD)

\$585

Library sales this month.

PENDING ORDERS

1

Orders waiting to be fulfilled.

Manage Catalog

Review and update the current library inventory.

View Full Inventory

Book Title	Category	Price	Stock	Actions
বিলিয়ন ডলার মুসলিম	HISTORY	\$150.00	10 in stock	Edit
মুমিনের ক্যারিয়ার ভাবনা	SELF_HELP	\$240.00	5 in stock	Edit
শিক্ষার্থীদের ক্যারিয়ার ভাবনা	SELF_HELP	\$240.00	10 in stock	Edit
ইতিহাসের আয়নায় বর্তমান বিশ্বব্যবস্থা	HISTORY	\$414.00	10 in stock	Edit

Order Management

Monitor and fulfill recent customer requests.

Live

GL-847251

PENDING

admin

5/14/2026

\$45.00

Details

GL-803087

DELIVERED

admin

5/14/2026

\$540.00

Details

Quick Insights

Inventory levels are stable. Focus on replenishing low stock products and keep the latest launches featured.

- Reorder top-selling titles weekly.
- Review pending delivery issues daily.
- Launch promotional bundles for loyal users.

Orders Page

BookVerse

Admin Dashboard

Sign Out

Analytics

Inventory

Orders

Customers

ADMIN / ORDER TRACKING

Customer Orders

Export CSV

Review and manage recent purchase rituals from your community.

TOTAL ORDERS

2

+12% vs last month

PENDING SHIPMENT

1

Requires attention

COMPLETED RITUALS

1

92.6% fulfillment rate

TOTAL REVENUE

\$585

Gross curated sales

All Orders

Pending

Shipped

Delivered

Search customer or ID...

ORDER ID	CUSTOMER NAME	DATE	TOTAL	STATUS	ACTIONS
GL-847251	 admin	5/14/2026	\$45.00	PENDING	
GL-803087	 admin	5/14/2026	\$540.00	DELIVERED	

Showing 1-2 of 2 results

< 1 2 3 >

BookVerse

HOME CATALOG FICTION PHILOSOPHY POETRY HISTORY DASHBOARD



Your Cart

(Total items: 4 items)



শিক্ষার্থীদের ক্যারিয়ার ভাবনা

- 2 +

\$480.00
\$240 each



বিলিয়ন ডলার মুসলিম

- 3 +

\$450.00
\$150 each



মুমিনের ক্যারিয়ার ভাবনা

- 2 +

\$480.00
\$240 each



ইতিহাসের আয়নায় বর্তমান বিশ্বব্যবস্থা

- 2 +

\$828.00
\$414 each

Order Summary

Subtotal

\$2238.00

Shipping

Free

Total

\$2238.00

PROMO CODE

GLOW15, RITUAL20...

Apply

Payment Method

Card

PayPal

COD

Shipping Details

Street address

City

Postal Code

Country

Pay \$2238.00

BookVerse

Curated literary experiences for readers who value timeless storytelling, elegant design, and the quiet beauty of the written word.

"A room without books is like a body without a soul."

CICERO

THE PRESS

Sustainability

Ingredients Glossary

Shipping & Returns

SUPPORT

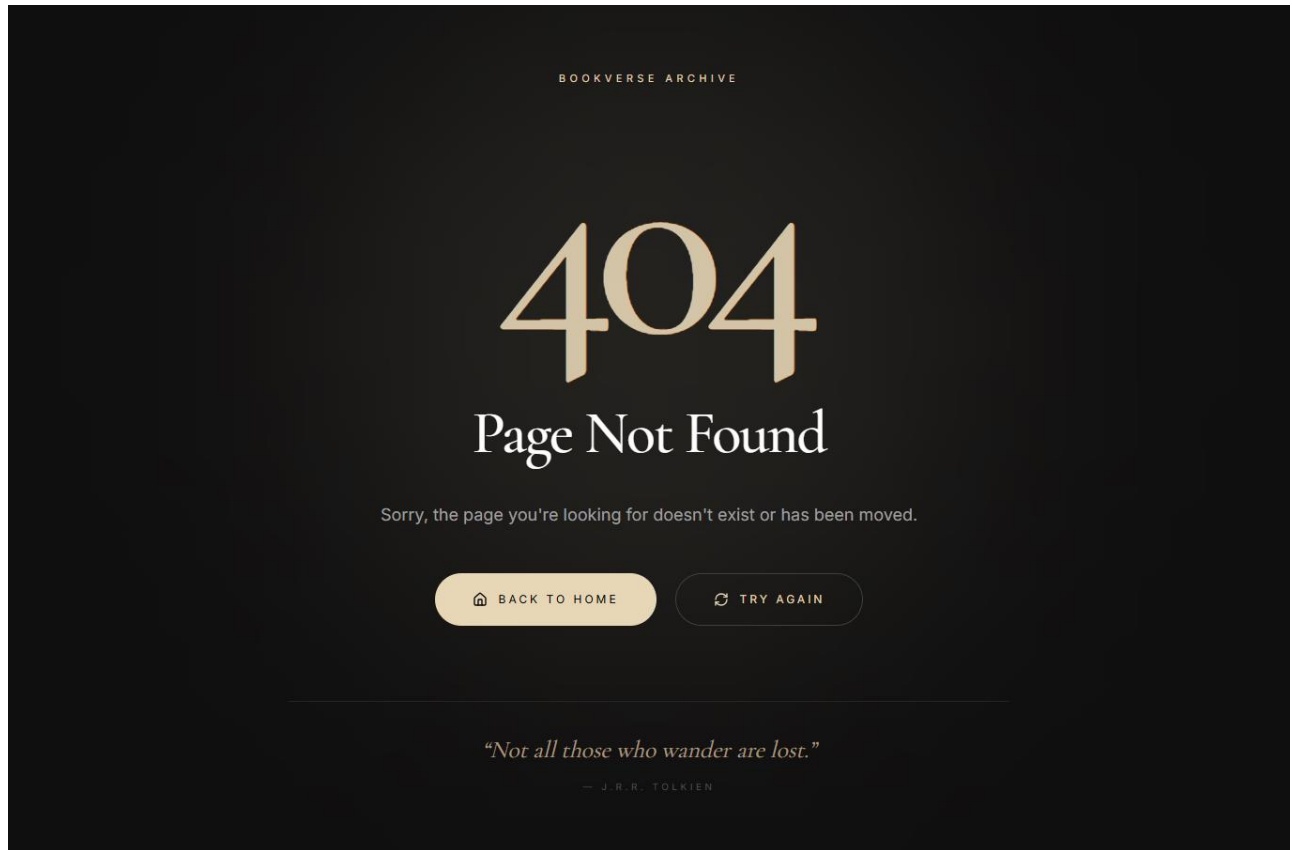
Privacy Policy

Contact

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INSTAGRAM TWITTER PINTEREST

Error Page



15. Future Improvements

Future enhancements planned for BookVerse include:

- Online Payment Gateway
- AI Book Recommendations
- Dark Mode
- E-book Reader
- Multi-language Support
- Advanced Analytics
- Wishlist Feature
- Reading Progress Sync
- Push Notifications
- Social Reading Features

16. Key Achievements of the BookVerse Project

1. Full-Stack MERN Application Successfully Developed

- A complete full-stack online bookstore platform has been successfully developed using MongoDB, Express.js, React.js, and Node.js.
- Frontend and backend systems have been fully integrated through RESTful APIs.

- Real-world eCommerce architecture and scalable development practices have been followed.

2. Secure Authentication & Authorization System

- A JWT-based secure authentication system has been implemented.
- Protected routes and private dashboard access have been successfully developed.
- Role-based authorization for users and admins has been implemented efficiently.
- Secure login, registration, and logout functionalities have been integrated.

3. Complete Online Bookstore Functionality

- Dynamic book browsing, filtering, searching, and categorization systems have been developed.
- Cart management, order placement, and order tracking systems have been implemented.
- A real-world online bookstore experience has been simulated successfully.

4. Dynamic User Dashboard Development

- Personalized reading dashboard functionality has been developed.
- Users can view reading progress, recent orders, reviews, and book collections dynamically.
- Interactive dashboard UI inspired by premium literary platforms has been designed.

5. Scalable Backend Architecture

- A modular MVC-inspired backend structure has been implemented.
- Controllers, routes, models, middleware, and services have been separated professionally.
- The backend architecture has been designed for future scalability and maintainability.

6. Responsive & Premium UI/UX Design

- A fully responsive UI has been developed for mobile, tablet, and desktop devices.
- Editorial-inspired typography and luxury book-focused layouts have been implemented.
- Tailwind CSS and DaisyUI have been used to create a modern and elegant user experience.

7. Real-Time API Integration

- Smooth frontend-backend communication has been implemented using Axios and React Query.
- Dynamic data fetching, caching, and synchronization systems have been integrated.
- Real-time catalog updates and dashboard rendering have been optimized.

8. Advanced Database Design & Management

- Structured MongoDB collections such as Users, Books, Orders, Reviews, and Cart have been designed efficiently.
- Data relationships and query optimization techniques have been implemented properly.
- MongoDB Atlas cloud database has been integrated successfully.

9. Security Best Practices Applied

- Password hashing using bcrypt has been implemented.
- JWT token-based protected routes have been secured.
- CORS configuration and secure API handling practices have been followed.
- Input validation and authentication middleware have been integrated.

10. Professional Frontend Engineering

- Reusable React components and clean folder structures have been maintained.
- React Router DOM has been used for dynamic routing and protected navigation.
- Modern React development practices and state management patterns have been followed.

11. Deployment & Production Readiness

- Frontend deployment compatibility with Netlify has been prepared.
- Backend APIs have been optimized for cloud hosting environments.
- Environment variable management and production-ready configuration have been maintained.

12. Real-World Project Experience & Collaboration

- Real-world full-stack project development workflow has been followed.
- Git and GitHub version control practices have been used throughout the project.
- Professional software engineering structure and scalable development methodology have been practiced successfully.

17. Conclusion

BookVerse successfully demonstrates a complete MERN stack application with modern frontend design, scalable backend APIs, secure authentication systems, and responsive UI/UX.

The project improved practical knowledge of:

- Full-stack web development
- REST API integration
- Modern React architecture
- Database management
- Authentication systems
- Responsive design
- Scalable project structure

BookVerse represents a professional real-world online bookstore platform built with modern technologies and clean development practices.

★★★★★ **END OF BOOKVERSE PROJECT REPORT** ★★★★★